Credit Risk Analytics Mini-Model

PD Estimation, Model Validation, IFRS 9-style ECL and Turkey Macro-Sector Overlay

Credit Risk Analytics / Risk Methodologies Portfolio Case

Enes Sarı

Data through: 8 May 2026 | Final revision date: 19 May 2026

Metric	Latest reading	Interpretation
Clean modeling observations	1,369,164	Resolved LendingClub modeling sample
Observed default rate	21.23%	Bad-rate base for PD modeling
Logistic Regression test AUC	0.709	Primary interpretable PD model
XGBoost challenger test AUC	0.718	Non-linear benchmark
Very High Risk default rate	38.86%	Risk-band separation signal
Very High Risk ECL share	53.05%	Expected-loss concentration
Base scenario ECL	USD 1.34bn	Illustrative ECL on test sample
Severe stress ECL increase	61.43%	Stress sensitivity
PD score PSI	0.00002 (Stable)	Train-test score stability
Latest Turkey total NPL ratio	2.78%	Local sector overlay

Table 1. Executive credit risk analytics snapshot.

Scope note: This memo is a portfolio case study. It uses the public LendingClub accepted-loan file family (accepted_2007_to_2018Q4.csv.gz) for the core PD/ECL workflow and public BDDK/TCMB indicators only as a separate Turkey macro-sector overlay.

1. Executive Summary

This case study builds a public-data credit risk analytics workflow using LendingClub loan-level data. The project develops an interpretable probability-of-default (PD) model, validates the model through out-of-sample performance and calibration checks, segments borrowers into risk bands, and translates predicted PDs into an illustrative IFRS 9-style expected credit loss (ECL) framework.

The primary logistic regression model achieved a test AUC of 0.709 and a test Gini of 0.417, while the XGBoost challenger model achieved a moderately higher test AUC of 0.718. Logistic regression is retained as the primary model because its predicted PDs are well-calibrated across deciles, and its drivers are easier to interpret in credit-risk governance terms.

The strongest portfolio finding is concentration. The Very High Risk band accounts for 31.2% of test-sample exposure but 45.7% of observed defaults and 53.1% of illustrative ECL. Under the severe stress scenario, total illustrative ECL increases from USD 1.34bn to USD 2.17bn, a 61.4% increase relative to the base case.

The project also adds model-stability and strategy layers: train-test PD score PSI is approximately 0.00002, indicating no material distribution shift; threshold simulation shows the trade-off between approval volume, retained exposure and default capture; and stress migration illustrates how exposure shifts toward higher-risk bands under adverse PD multipliers. A separate Turkey macro-sector overlay uses BDDK and TCMB indicators to show how similar monitoring logic can be connected to local banking-sector conditions.

2. Dataset and Default Definition

The core modeling dataset uses the public LendingClub accepted-loan file family, accepted_2007_to_2018Q4.csv.gz. To approximate an application-time PD model, the variable set is restricted to information available at or near origination, such as loan amount, funded amount, term, interest rate, installment, grade, employment length, home ownership, annual income, debt-to-income ratio, FICO proxy, delinquencies, inquiries, revolving utilization, public records and loan purpose. Post-origination variables such as recoveries, total payments and last payment information are excluded to reduce target leakage.

The binary default flag is based on resolved or materially default-like loan statuses. Fully paid loans are treated as non-default observations. Charged-off, defaulted and late 31-120 day loans are treated as default observations. Current, issued, grace-period and short-term delinquency observations are excluded from the binary modeling sample.

Target class	LendingClub status mapping	Modeling treatment
Good / non-default	Fully Paid; Does not meet the credit policy. Status:Fully Paid	default_flag = 0
Bad / default	Charged Off; Default; Late (31-120 days); not meet the credit policy. Status: Charged Off	default_flag = 1
Excluded	Current; Issued; In Grace Period; short-term delinquency or unresolved performance states	Excluded modeling sample

Default definition box. Binary target construction and status treatment.

Item	Reading	Interpretation
Raw selected observations	2,260,701	Selected LendingClub columns read from raw file
Clean modeling observations	1,369,164	After status filter and core missing-value checks
Observed default rate	21.23%	Charged-off/default/late 31-120 definition
Total EAD proxy in clean data	USD 19.77bn	Funded amount used as exposure proxy

Table 2. Modeling data construction summary.

3. PD Model Development

The project estimates two models. Logistic regression is used as the primary PD model because it is transparent and aligned with traditional credit scorecard logic. XGBoost is trained as a challenger model to test whether a non-linear structure improves discrimination. Both models are trained on the same leakage-controlled feature set and evaluated on a 30% holdout sample using stratified splitting.

An initial class-weighted logistic specification was rejected because it produced overstated PD levels despite improving default recall. The final logistic model is estimated without class weighting to prioritize probability calibration, which is more appropriate for PD estimation and ECL translation than a pure classification objective. A compact driver table is added below to make the expected feature-risk relationships visible to non-technical readers and model reviewers.

Driver	Expected risk direction	Model interpretation
Higher interest rate	Higher PD	Pricing embeds borrower risk
Lower FICO proxy	Higher PD	Weaker credit quality
Higher DTI	Higher PD	Debt burden risk
More delinquencies	Higher PD	Prior repayment stress
Longer term	Higher PD	Longer exposure horizon
Higher revolving utilization	Higher PD	Liquidity pressure

Model	Sample	AUC	Gini	Precision	Recall
Logistic Regression	Train	0.710	0.420	0.558	0.070
Logistic Regression	Test	0.709	0.417	0.560	0.070
XGBoost	Train	0.722	0.444	0.579	0.087
XGBoost	Test	0.718	0.436	0.575	0.085

Table 3. Model performance summary.

Classification metrics use a 0.50 threshold for transparency. The model is primarily used for PD estimation, calibration, ranking and risk segmentation rather than binary default classification at a fixed threshold. Because the model is used for PD ranking and ECL translation rather than hard binary classification, AUC, Gini, calibration and risk-band separation are more decision-relevant than recall at an arbitrary 0.50 threshold.

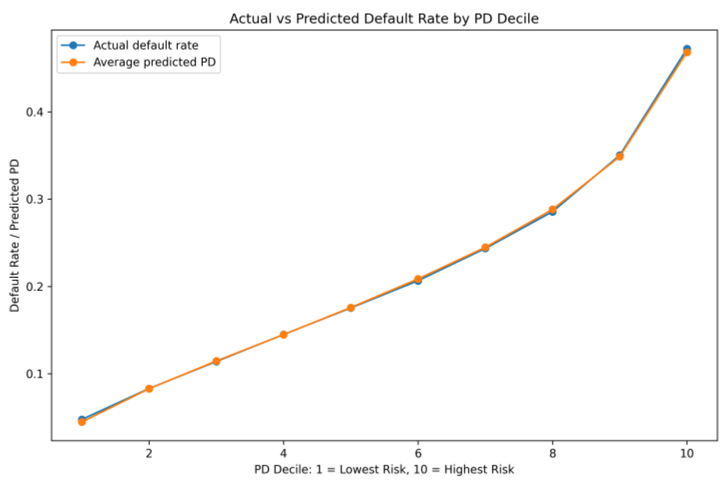


Figure 1. Actual versus predicted default rate by PD decile.

4. Model Validation and Calibration

Model validation focuses on both discrimination and calibration. Discrimination is measured through ROC/AUC and Gini. Calibration is evaluated by comparing average predicted PDs with realized default rates by PD decile. The final logistic model shows strong out-of-sample calibration: the first decile has a realized default rate of 4.76% versus an average predicted PD of 4.47%, while the tenth decile has a realized default rate of 47.24% versus an average predicted PD of 46.85%.

This decile-level behavior supports the use of the logistic model as the primary PD source for risk bands and illustrative ECL. XGBoost produces a modestly higher AUC, but the logistic model is preferred for the main risk workflow because its probability outputs are easier to validate and explain.

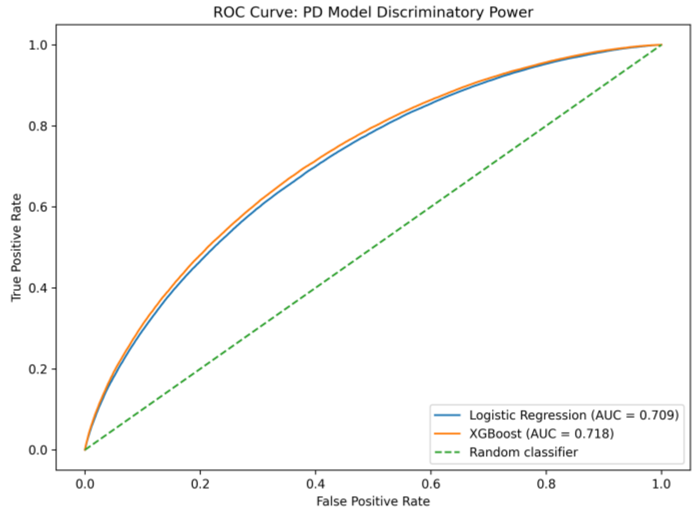


Figure 2. ROC curve and discriminatory power of the logistic and XGBoost PD models.

5. Risk Segmentation

Borrowers are segmented into four quantile-based risk bands using the primary logistic PD. The bands show a clean monotonic default gradient: Low Risk loans have a 7.37% realized default rate, while Very High Risk loans have a 38.86% realized default rate. The Very High Risk band accounts for 31.2% of exposure but 45.7% of observed defaults, indicating a material credit risk concentration.

Risk band	Loans	Exposure	Avg PD	Actual default	Exposure share	Default share
Low Risk	102,688	USD 1.36bn	7.2%	7.4%	23.0%	8.7%
Medium Risk	102,687	USD 1.31bn	15.3%	15.2%	22.1%	17.9%
High Risk	102,687	USD 1.40bn	23.7%	23.5%	23.7%	27.6%
Very High Risk	102,688	USD 1.85bn	38.7%	38.9%	31.2%	45.7%

Table 4. Risk-band summary based on primary logistic PD.

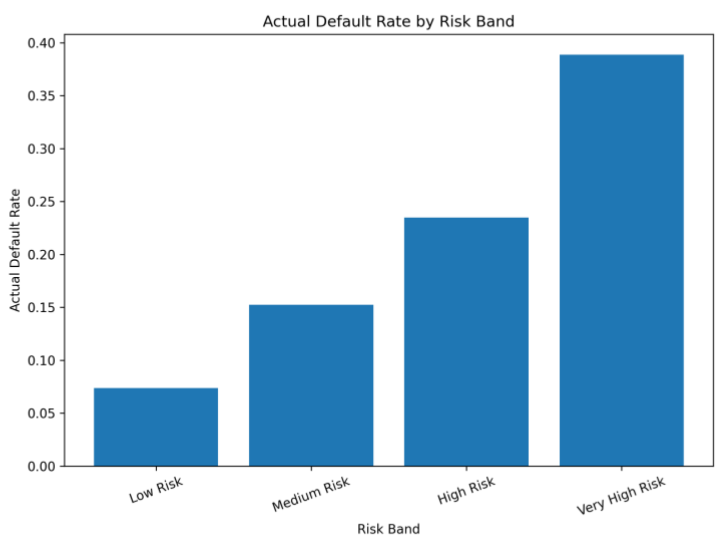


Figure 3. Actual default rate by risk band.

6. IFRS 9-style ECL Framework

The ECL layer translates calibrated PDs into a simplified expected-loss framework. EAD is proxied using funded loan amount. LGD is assumed at 45% in the base case because the public dataset does not provide a reliable internal recovery, collateral and workout dataset. Stage 1 consists of non-defaulted Low and Medium Risk loans; Stage 2 consists of non-defaulted High and Very High Risk loans; and Stage 3 consists of observed defaulted loans. This is an illustrative IFRS 9-style approximation rather than a regulatory implementation. In a production IFRS 9 framework, Stage 2 assignment would require comparing current lifetime risk against origination risk and applying bank-specific significant-increase-in-credit-risk (SICR) rules; here, high predicted PD is used only as a transparent public-data proxy.

Stage proxy	Loans	Exposure	Avg ECL PD	ECL	Exposure share	ECL share	ECL rate
Stage 1	182,171	USD 2.38bn	11.0%	USD 117mn	40.0%	8.7%	4.9%
Stage 2	141,359	USD 2.19bn	59.3%	USD 611mn	37.0%	45.5%	27.9%
Stage 3	87,220	USD 1.36bn	100.0%	USD 614mn	23.0%	45.8%	45.0%

Table 5. Illustrative IFRS 9-style stage and ECL summary.

Stage 2 and Stage 3 together account for 91.3% of illustrative ECL, despite representing a smaller share of exposure than the full portfolio. This reinforces the importance of credit monitoring and risk migration analysis in expected-loss management.

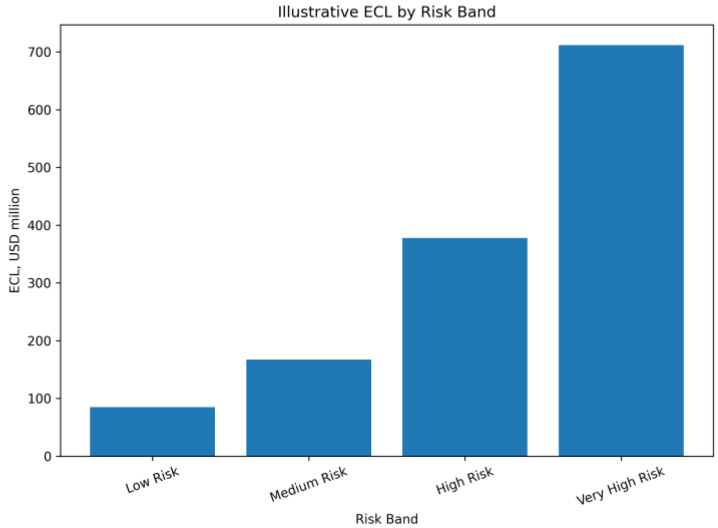


Figure 4. Illustrative ECL by risk band.

7. Stress Testing and Portfolio Impact

The stress framework applies scenario-level PD multipliers and LGD assumptions. The base case uses the calibrated ECL PD and a 45% LGD. Mild deterioration, adverse, and severe stress scenarios increase both PD and LGD. EAD is held constant to isolate credit-quality deterioration rather than balance-sheet growth effects.

Scenario	PD multiplier	LGD	Total ECL	ECL / exposure	Increase vs base
Base	1.00x	45.0%	USD 1.34bn	22.6%	0.0%
Mild Deterioration	1.15x	50.0%	USD 1.60bn	26.9%	19.2%
Adverse	1.25x	55.0%	USD 1.83bn	30.9%	36.5%
Severe Stress	1.50x	60.0%	USD 2.17bn	36.5%	61.4%

Table 6. Scenario ECL summary.

Under the severe stress scenario, total illustrative ECL rises from USD 1.34bn to USD 2.17bn, a 61.4% increase versus the base case. The result shows how a relatively simple PD-LGD-EAD engine can be used for management-style sensitivity analysis.

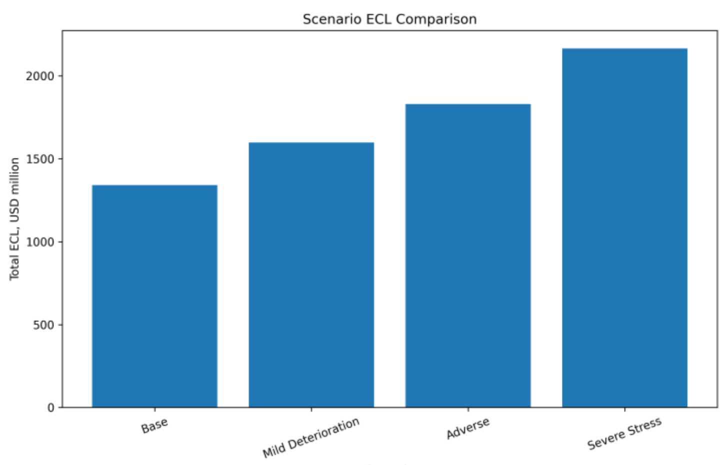


Figure 5. Scenario ECL comparison.

8. Model Stability and Strategy Simulation

The train-test PD score PSI is approximately 0.00002, which indicates no material distribution shift between the model development and holdout samples. This supports the stability of the score distribution in the holdout validation setup.

The threshold simulation is not an underwriting recommendation. It illustrates how a PD model can support risk strategy discussions by showing the trade-off between approval rate, retained exposure and the share of defaults captured in a review or rejection population. At a 20% PD threshold, the model approves 52.5% of loans, retains 47.3% of exposure and captures 71.1% of defaults in the rejected or manual-review segment.

PD threshold	Approval rate	Approved default rate	Exposure retained	Default capture in review/reject
10%	20.2%	6.6%	18.8%	93.7%
20%	52.5%	11.7%	47.3%	71.1%
30%	77.5%	15.8%	71.4%	42.4%
40%	90.9%	18.5%	87.3%	20.7%
50%	97.3%	20.3%	96.1%	7.0%

Table 7. Approval-threshold simulation summary.

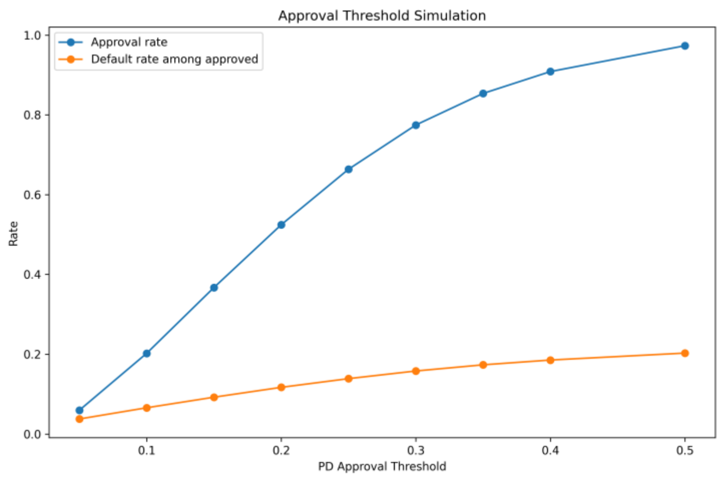


Figure 6. Approval threshold simulation.

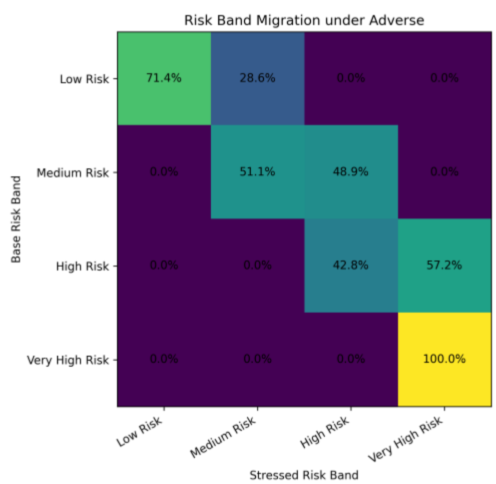


Figure 7. Risk band migration under adverse stress.

9. Turkey Macro-Sector Overlay

The Turkey overlay is included to connect the credit risk workflow to local banking-sector monitoring. It is not used as a model input because the core PD model is estimated on U.S. LendingClub borrower-level data. Instead, public BDDK loan and NPL indicators and TCMB macro variables are used as contextual indicators for how a similar framework would be read in a Turkish banking environment. The overlay demonstrates how the same PD/ECL monitoring logic could be contextualized for a Turkish banking portfolio if institution-level borrower data were available.

Indicator	Value
Latest selected date	2026-05-08
Latest total NPL ratio	2.78%

Latest consumer NPL ratio	4.63%
Latest commercial NPL ratio	2.18%
Latest SME NPL ratio	3.61%
Latest coverage ratio	75.42%
Latest policy rate	37.00%
Latest CPI YoY	32.37%
Total loan growth since first selected date	41.93%
NPL growth since first selected date	157.20%

Table 8. Turkey macro-sector overlay snapshot.

The selected Turkey overlay shows that total loans increased by 41.9% while NPL stock increased by 157.2% between the first selected date and 8 May 2026. The latest total NPL ratio is 2.78%, while the consumer NPL ratio is higher at 4.63%. This supports the relevance of portfolio monitoring, risk segmentation, stress testing, and ECL-style analysis for local credit-risk functions.

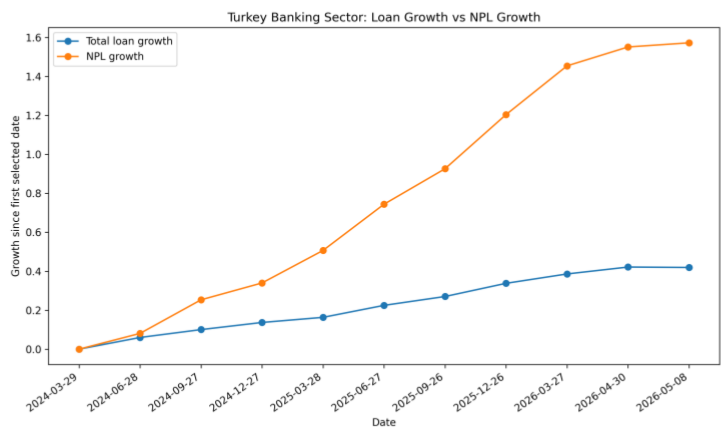


Figure 8. Turkey banking-sector loan growth versus NPL growth.

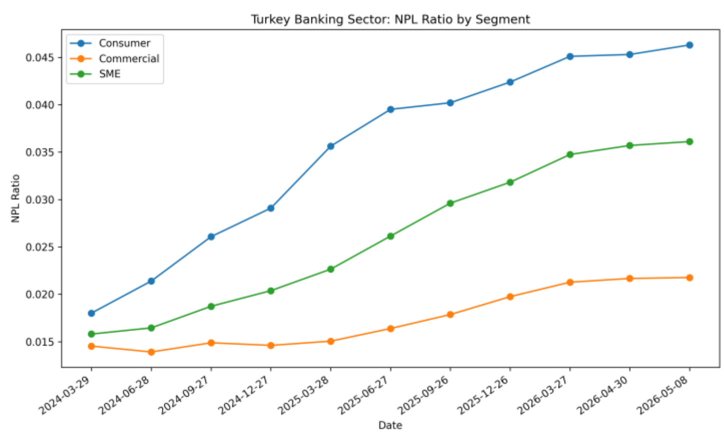


Figure 9. Turkey banking-sector NPL ratios by segment.

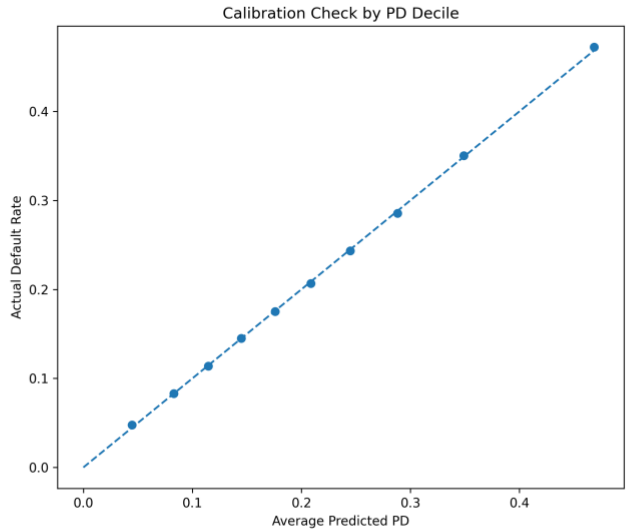
10. Conclusion and Application Relevance

The case demonstrates a practical credit risk analytics workflow: define a default outcome, build a leakage-controlled modeling dataset, estimate an interpretable PD model, benchmark against a challenger algorithm, validate discrimination and calibration, translate PDs into risk bands and illustrative ECL, and extend the model output into stress, strategy, and monitoring views.

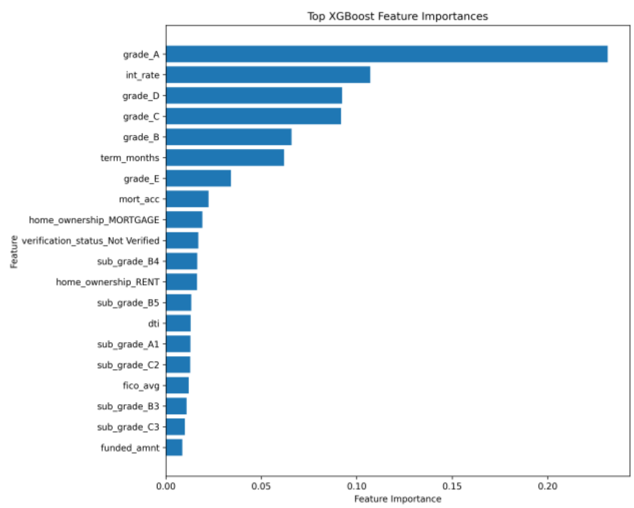
The strongest application signal is the combination of calibrated PD modeling, transparent validation, expected-loss translation, portfolio concentration analysis, and management-style reporting. The project is deliberately conservative about what cannot be inferred from public data. It does not claim to reproduce a bank's internal IFRS 9 model, internal ratings-based model, production underwriting engine, or true regulatory ECL framework.

For credit risk analysts, risk methodologies, model validation, IFRS 9 analytics, credit portfolio monitoring, and financial services advisory roles, the project demonstrates how quantitative modeling and reporting skills can be applied to banking risk use cases with public data and reproducible Python workflows.

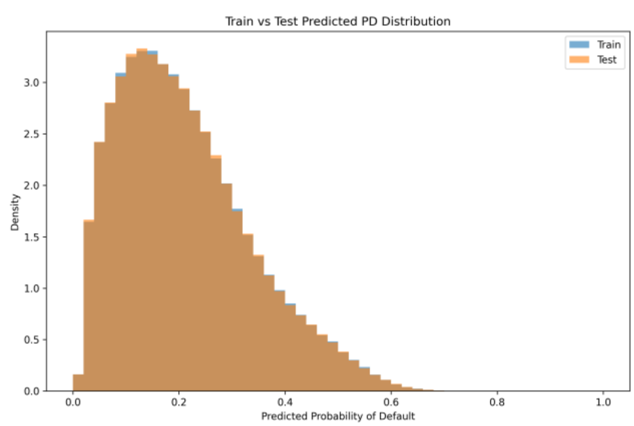
Appendix A - Additional Model Outputs



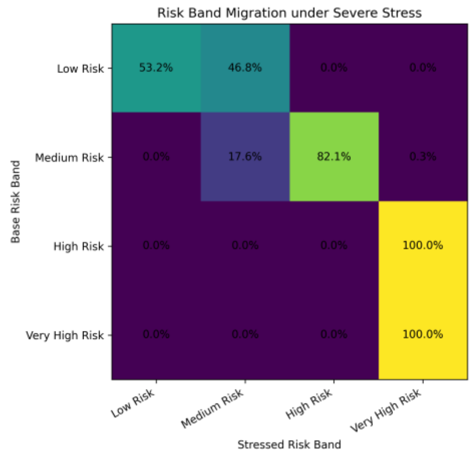
Appendix Figure A1. Calibration check by PD decile.



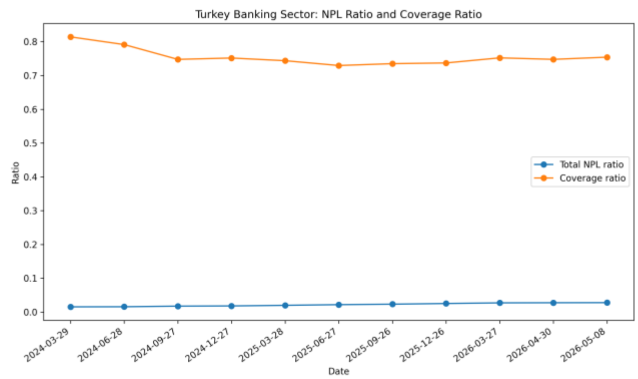
Appendix Figure A2. XGBoost challenger feature importance.



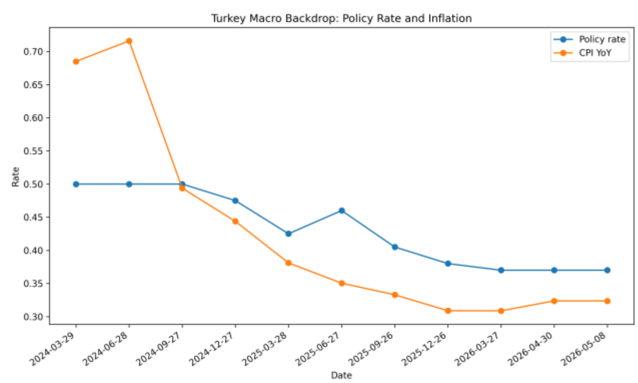
Appendix Figure A3. Train versus test predicted PD distribution.



Appendix Figure A4. Risk band migration under severe stress.



Appendix Figure A5. Turkey banking-sector total NPL ratio and coverage ratio.



Appendix Figure A6. Turkey macro backdrop: policy rate and inflation.

Appendix B - Methodology and Caveats

Component	Construction	Caveat
PD model	Logistic regression primary model; XGBoost challenger	Public loan-level data; not a production underwriting model
Default definition	Charged-off, default and late 31-120 day loans treated as bad	Resolved-loan sample; current loans excluded
EAD proxy	Funded amount	Simplified exposure-at-default proxy
LGD assumption	45% base; 50%-60% under stress	Assumed due to absence of internal recovery and collateral data
Stage proxy	Risk-band and observed-default based Stage 1/2/3 approximation	Illustrative IFRS 9-style framework only
PSI	Train-test PD score distribution comparison	Stability checks within the sampled holdout design
Turkey overlay	BDDK loan/NPL indicators and TCMB policy/inflation variables	Contextual only; not used as model features

Appendix Table B1. Methodology summary and caveats.